



TrioDocs

Version: 0.4.0

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Remote Control

Remote Control

The limited use of remote control with Nightscout, i.e., entry of Carb Correction and Temporary Targets when Careportal is authenticated, continues to be supported with Trio.

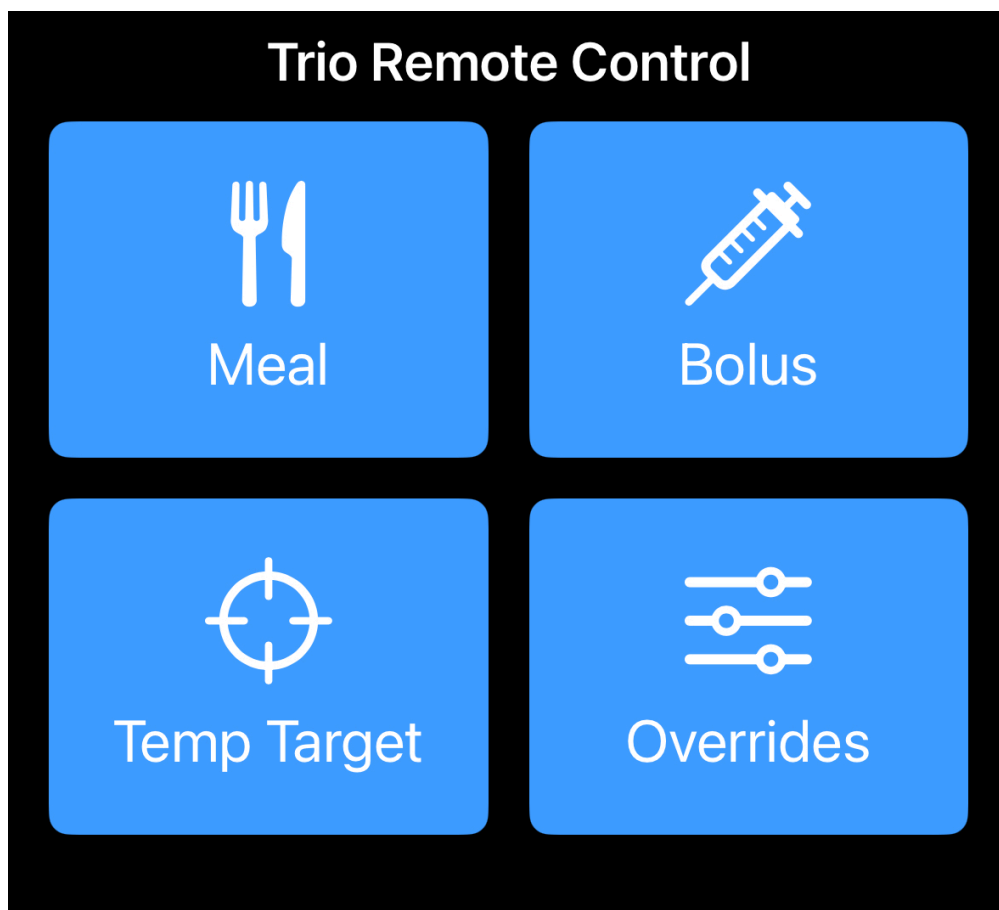
Nightscout version must be 15.0.2 or newer to properly display the OpenAPS pill with Trio 0.5.x (and newer)

Additional remote capabilities are offered for Trio using the *LoopFollow* app with these versions:

- Trio 0.5.x (and newer)
- *LoopFollow* version 2.4.0 (and newer)

The *LoopFollow* Trio Remote Control screen provides access to:

- Meal (Carbs or Carbs & Bolus)
- Bolus
- Temp Target
- Overrides



To ensure security, these commands are protected by a shared secret located on the Trio user's Remote Control menu screen, which must be entered in the *LoopFollow* app. The Trio app is the primary controller of whether remote commands are enabled or disabled.

? How does this differ from Trio 0.2.x?

Remote Control using Trio 0.5.x and newer (`dev` branch) is different from the version supplied by Trio 0.2.x and older.

🔥 Remote Control Changes if Updating from Trio 0.2.x

Trio can use Nightscout Careportal to enter `Carb Correction`, and start and cancel `Temporary Target`.

- This was available in Trio 0.2.x and continues to be available in Trio 0.5.x (and newer).

Trio 0.2.x supported other remote options (using announcements via Careportal).

- Those options were replaced by the more secure Trio Remote Control for Trio 0.5.x (or newer).
- Trio Remote Control requires the use of the *LoopFollow* app version 2.4.0 or newer
- **Using announcements to provide remote control of the Trio phone is no longer supported**

LoopFollow Overview

Experienced *LoopFollow* users should still read this section - it touches on important configuration requirements for remote control.

The graphic below shows the *LoopFollow* features on the main screen of the app.

Glucose

Glucose Trend

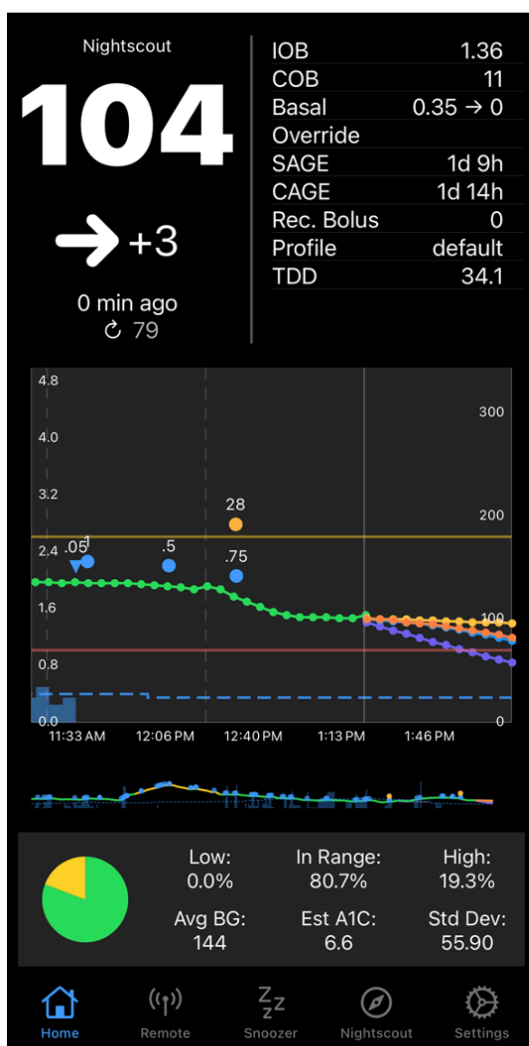
Last Loop Cycle
Eventual Glucose

Scrollable/scalable
graph display with
BG, IOB, COB,
Basal, Bolus, and
the General
Nightscout
Careportal info

24-hour graph

24-hour results

Toolbar



Information Display Settings

IOB	☒
COB	☒
Basal	☒
Override	☒
Battery	☐
Pump	☐
SAGE	☒
CAGE	☒
Rec. Bolus	☒
Min/Max	☐
Carbs today	☐
Autosens	☒
Profile	☒
Target	☐
ISF	☐
CR	☐
Updated	☐
TDD	☒
IAGE	☐

LoopFollow Settings

You tap on the settings icon at the bottom right of the toolbar to configure *LoopFollow*. The setting screen is shown in the graphic below.

DATA SETTINGS

Units

mg/dL

mmol/L

Nightscout Settings



Dexcom Settings



APP SETTINGS

Background Refresh Settings



General Settings



Graph Settings



Information Display Settings



Alarms



Remote Settings



INTEGRATIONS

Calendar



Contact



ADVANCED SETTINGS

Advanced Settings



LOGGING

View Log



[Share Logs](#)

BUILD INFORMATION

Version

2.4.0



Home



Remote



Snoozer



Nightscout



Settings

Add Data Source for *LoopFollow*

You provide *LoopFollow* with information about the person you are following. At least one of these must be entered:

- [Nightscout URL with access token](#)
- [Dexcom Share credentials](#)

Add Nightscout

The [Nightscout URL](#) with an appropriate token is required:

- To enable display of the Information Table
- To configure remote control using `Nightscout` or `Trio Remote Control`
 - `Nightscout` requires a minimum token of `readable` and `careportal`
 - `Trio Remote Control` requires a minimum token of `readable`

The graphic below shows the Data Settings display when you tap on the [Nightscout Settings](#) row of *LoopFollow* Settings.

To simplify setup, you can copy your [Nightscout URL](#) (including the token) from the [Admin Tools in Nightscout](#). When pasted into *LoopFollow* [URL](#) row, the app will automatically extract and fill in both the [URL](#) and token.

Nightscout Settings Done

URL

URL Enter URL with Token

TOKEN

Token

STATUS

Address Empty Confirm access

Nightscout Status

Wait a few seconds after entering the [URL](#) to observe the Status. As long as you see `OK`, the Information Table is displayed.

Status	(Token Type)	Comment
--------	--------------	---------

OK	(Admin)	Works for any kind of remote control
OK	(Read & Write)	Required for <code>Nightscout</code> remote control Works for any kind of remote control
OK	(Read)	Required for <code>Trio Remote Control</code>

To use Trio Remote Control, your `Nightscout` authentication token only needs read access, but you must have `Apple Push Notification System (APNS)` keys configured.

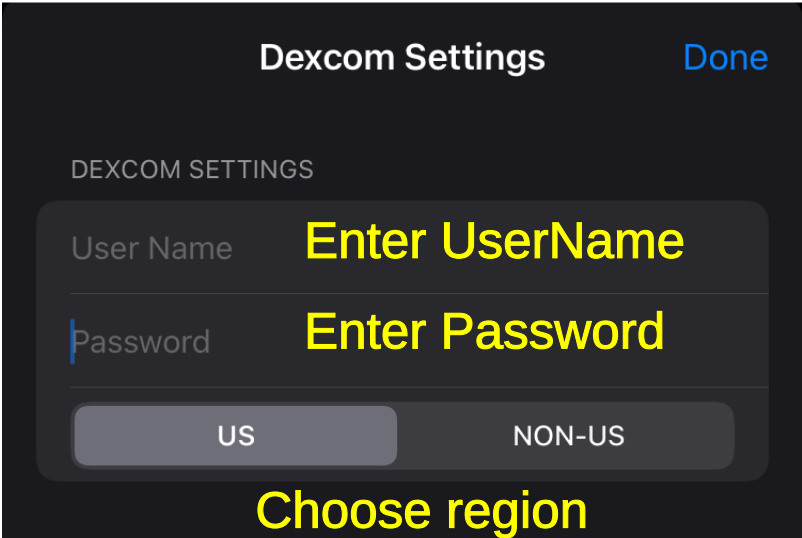
Potential Status Error messages:

- Invalid `URL`
- Network Error
- Invalid Token
- Token Required
- Site Not Found
- Address Empty
- Unknown Error

Add Dexcom

The graphic below shows the Data Settings display when you tap on the Dexcom Settings row of `LoopFollow` Settings.

The Dexcom Share credentials are optional, but can be useful when the `Nightscout` URL is unavailable.



Trio App: Enable Remote Control

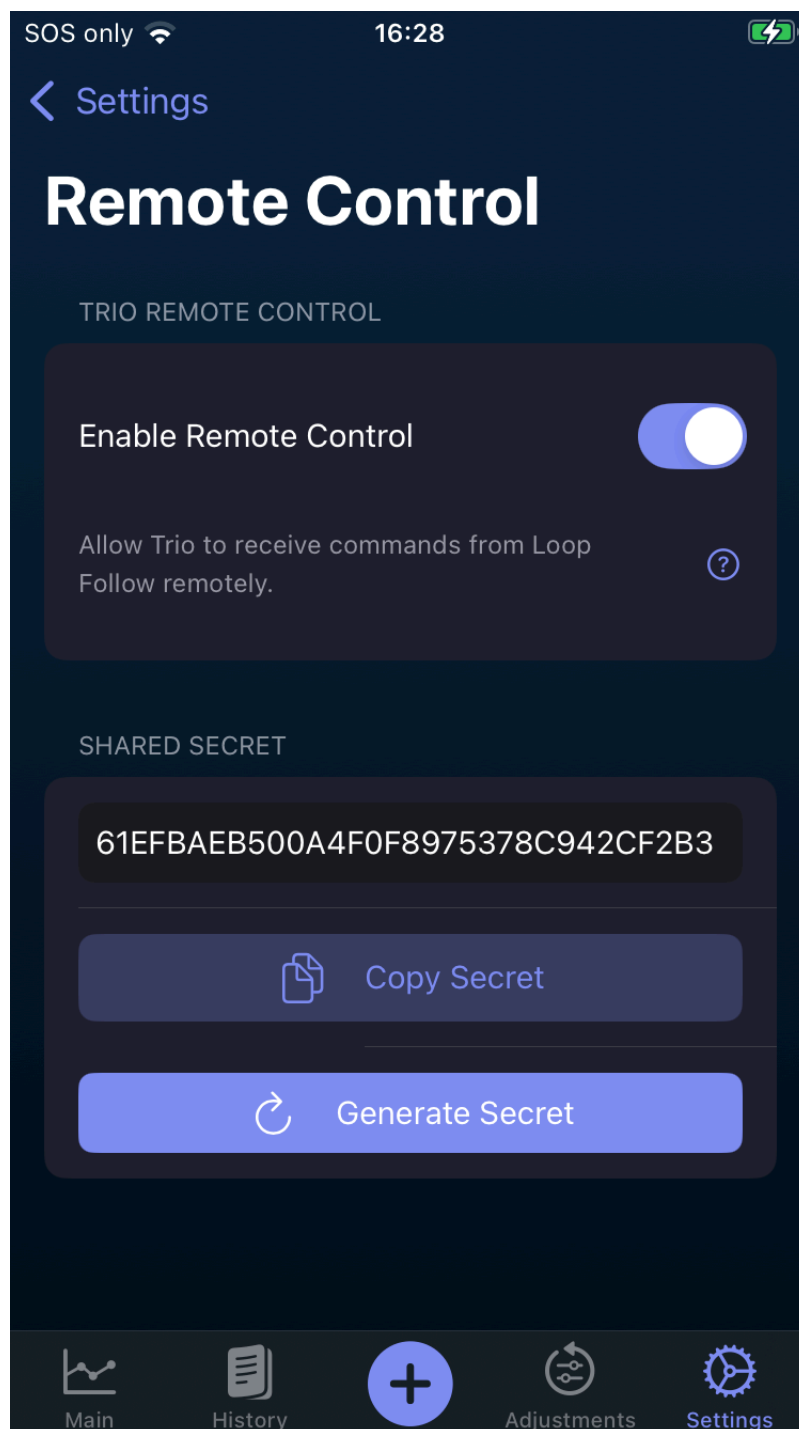
Default: `OFF`

When you configure remote control, you should have access to the *LoopFollow* phone and the *Trio* phone and test that it works as expected.

Remote control must be enabled on the Trio phone as well as configured on the *LoopFollow* phone. The Trio screen is shown below.

You can search for this screen in Trio settings or go through the sequence: Trio, Settings, Features, Remote Control.

The `SHARED SECRET` should be copied from the Trio phone and added to the `Shared Secret` row of the *LoopFollow* Remote Settings screen.



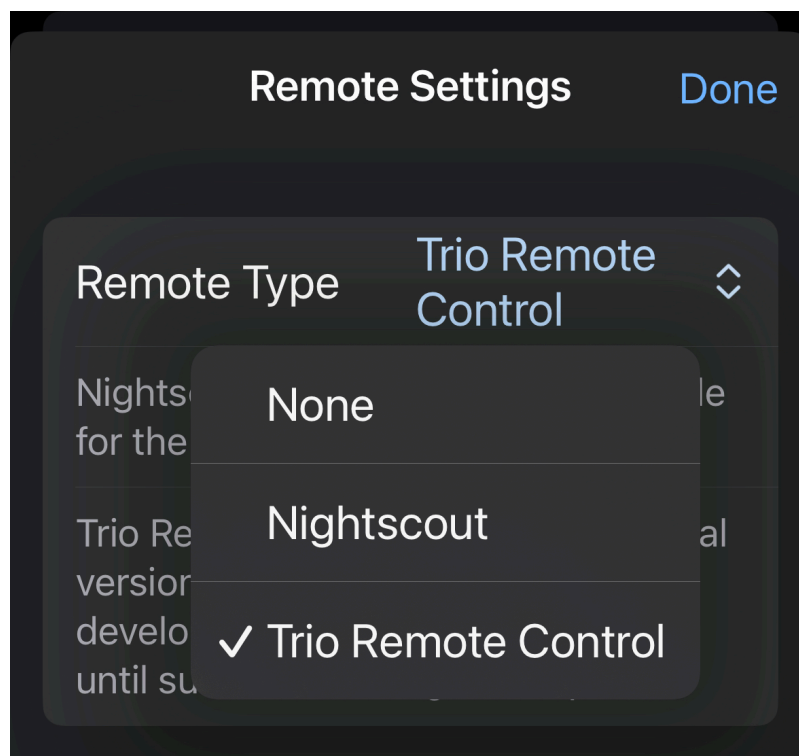
When Remote Control is enabled, you can send boluses, set overrides or temporary targets, add carbs, and other commands to Trio via push notifications.

Important

The ability for the Trio app to be remotely controlled will be **disabled** when `Enable Remote Control` is turned OFF, even if you have *LoopFollow* configured with the correct shared secret. This is for the protection of the Trio user, so that they **always** are the primary controller of their insulin dosing app.

Select Type of Remote Setting for *LoopFollow*

The Remote Settings row in the *LoopFollow* Settings screen is used to select the type of remote access you wish to use.



WARNING: The `Trio Remote Control` option is only available if you have already entered a [Nightscout URL](#) that is recognized as a Trio site.

- Trio Remote Control
 - Continue with [LoopFollow App: Trio Remote Control](#) to finish the configuration process
- [Nightscout](#)
 - Remote control is limited to starting and canceling Temp Targets
 - Your [Nightscout](#) must be configured for remote control as explained in [Nightscout for Temp Target Control](#)

Remote Options

The *LoopFollow* app provides amazing display and alarm capabilities, whether remote control is enabled or not.

It can be used with the Trio or the Loop app with the Nightscout option for starting and stopping temp targets (Trio) or overrides (Loop).

The graphic below shows the top portion of the Remote Settings screen when None, Nightscout or Trio Remote Control is selected. The lower portion of the screen is found in the Guardrails section.

SOS only

13:01



Remote Settings

Done

REMOTE TYPE

None



Nightscout

Trio Remote Control

Nightscout is the only option for Loop.

SOS only

13:01



Remote Settings

Done

REMOTE TYPE

None

Nightscout



Trio Remote Control

Nightscout is the only option for Loop.

USER INFORMATION

User

Enter User

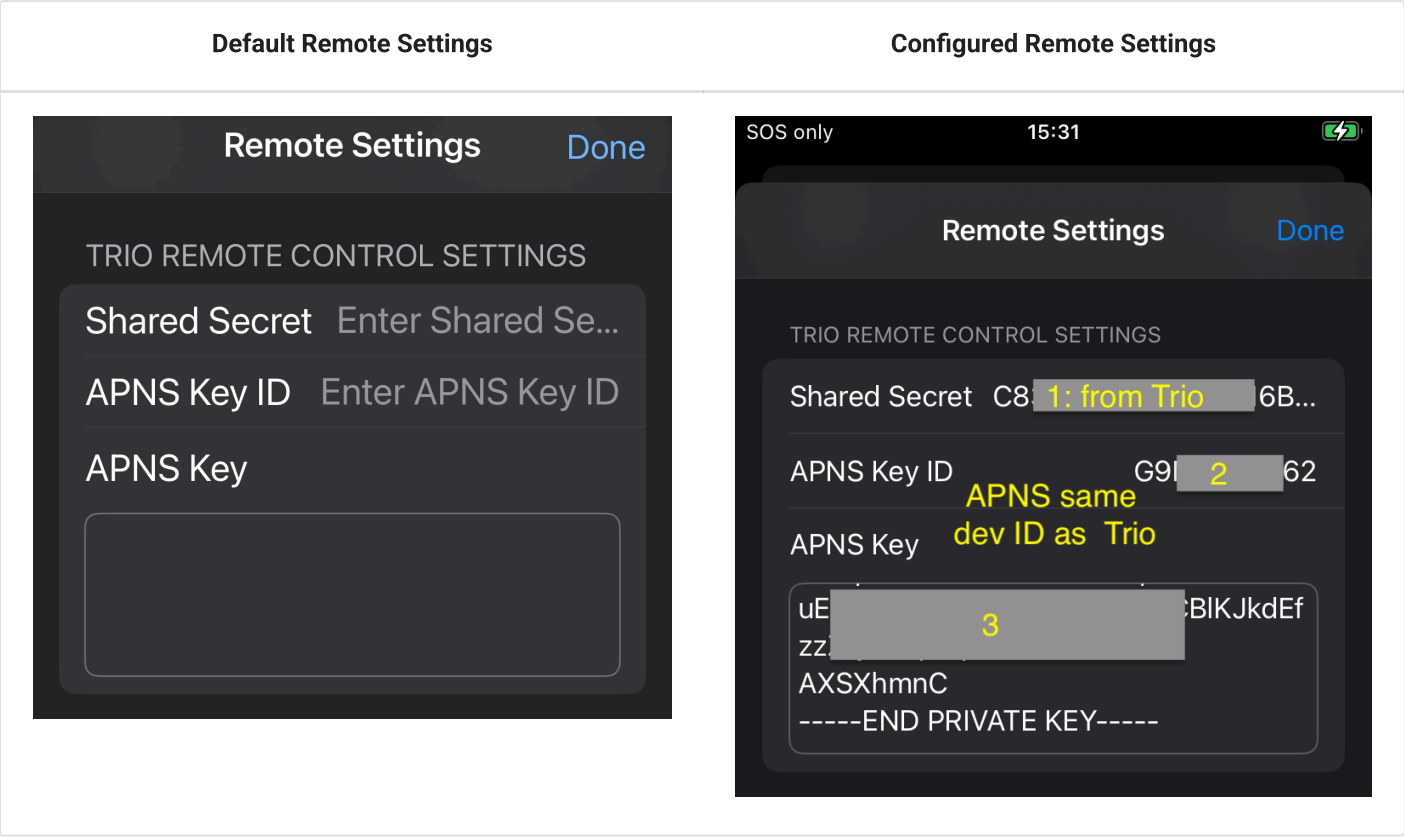
SOS only

13:02



LoopFollow App: Trio Remote Control

When you select Trio Remote Control as the Remote Type in the *LoopFollow* app, you must fill in the (1) [Shared Secret](#), (2) [APNS Key ID](#) and (3) [APNS Key](#).



User

The person using the *LoopFollow* app should enter the name they want to show up as having entered this entry.

- At the current time, this is not used by *LoopFollow* for Trio
- It does show up as a notification on a *Loop* phone when *LoopFollow* is used with the *Loop* app
 - This feature might be added to *LoopFollow* for Trio at a later time

Shared Secret

This is the unique shared secret that can be generated or entered into the Trio app in the Remote Control screen. The shared secret in Trio and *LoopFollow* must match to provide the ability to remotely send commands to this Trio app.

Please use a secure secret - the automatically generated secret is recommended.

APNS Key ID

See [Nightscout for Temp Target Control](#). The value of the `LOOP_APNS_KEY_ID` goes here.

APNS Key

See [Nightscout for Temp Target Control](#). The value of the `LOOP_APNS_KEY` goes here.

Guardrails

The maximum allowed entries for Bolus, Carbs, Protein, and Fat are configured in the guardrails section shown in the graphic below. This example is one in which the Shared Secret and APNS values have not yet been added.

GUARDRAILS

Max Bolus	1.00 U
Max Carbs	30 g
Max Protein	30 g
Max Fat	30 g

MEAL SETTINGS

Meal with Bolus	☒
Meal with Fat/Protein	☒

DEBUG / INFO

Device Token:
Production Env.: False
Team ID:
Bundle ID:

Meal Settings

The user can decide to enable or disable two features independently.

- Meal with Bolus
 - When enabled, a bolus command can be sent at the same time as the meal entry
- Meal with Fat/Protein
 - When enabled, the user is presented with a Protein and Fat row in addition to the Carbs and Bolus Amount rows

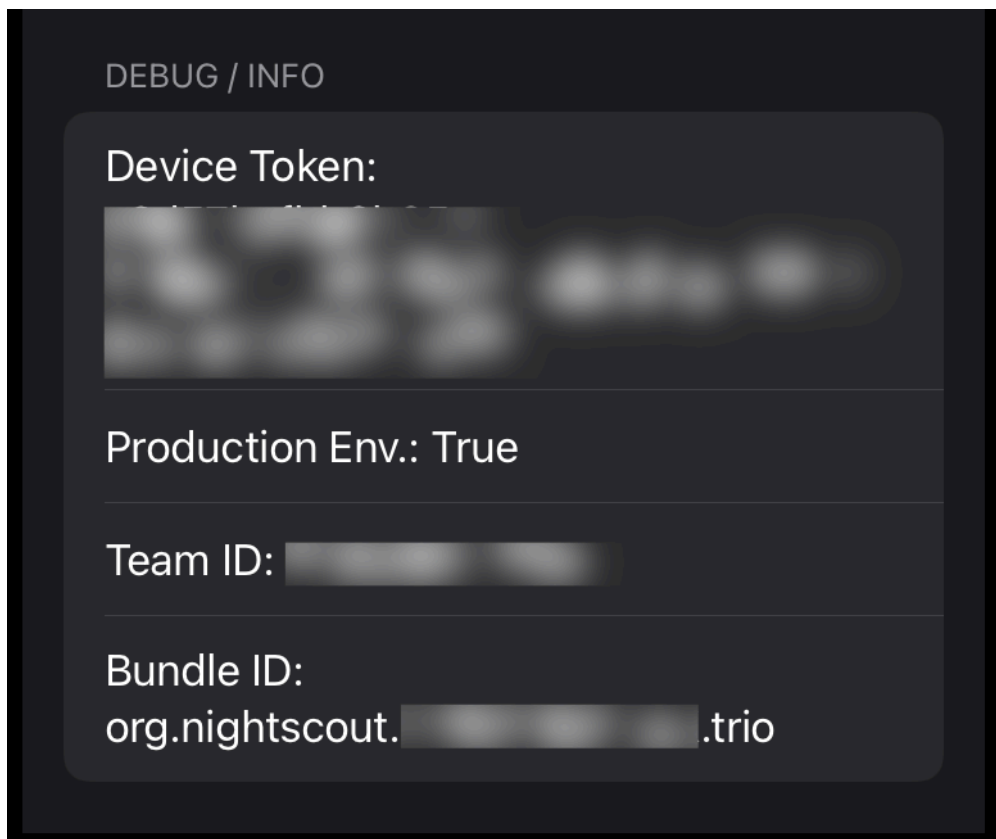
Refer to the graphic in the [Guardrails](#) section.

Debug / Info

This section indicates if the *LoopFollow* app entries match the Shared Secret in the Trio phone and contain the valid Apple Push Notification Values. If all rows are not filled out, the settings need to be updated. Return to [LoopFollow App: Trio Remote Control](#) and try again.

Refer to the graphic in the [Guardrails](#) section to view an example when the Shared Secret and APNS values have not been entered.

The graphic below shows a properly configured *LoopFollow* when the Trio app was built using the Browser Build method.



Using Trio Remote Control

Once the *LoopFollow* phone is configured, and while the Trio phone is handy, test sending Remote Commands. It is good to also have a browser open with the Nightscout URL displayed.

Remember to give the system time to update.

The sequence is *LoopFollow* to *Apple Push Notifications* to *Trio*, which uploads to *Nightscout* and then is displayed in the *LoopFollow* main screen.

Remote Meal

More info coming soon!

When entering meals and choosing to schedule the meal, any bolus included in the meal is enacted immediately. Only the carb entry is entered according to the schedule.

Network Provider

22:01



< Trio Remote Control

Meal

MEAL DATA

Carbs	3 g
Protein	0 g
Fat	0 g
Bolus Amount	1 U

SCHEDULE

Schedule for later



Select Time

22:01

Note: The carbs will be scheduled, but the bolus is enacted immediately.

Send Meal



Remote Bolus

More info coming soon!

Temp Target

More info coming soon!

Overrides

More info coming soon!

Nightscout for Temp Target Control

The configuration for Nightscout described in this section is required for either of these cases:

- You selected the remote type of `Nightscout` for your *LoopFollow* settings
- You want to use the Remote Options in the Careportal at your Nightscout site

If you are new to this topic, please refer to [LoopDocs: Remote Configuration](#) for more details

- The same set of `Nightscout` config vars is used by `Trio` and `Loop`
 - Ignore references to `LoopCaregiver` in the `LoopDocs` link, you will use `LoopFollow` or the `Nightscout Careportal`
- The Apple Push Notification (APNS) config vars must be configured for the *Apple* developer ID of the person who builds the Trio app (`LOOP_DEVELOPER_TEAM_ID` in the table below)
- The first two values in this table are the same ones added to the `LoopFollow` app when configuring the [LoopFollow App: Trio Remote Control](#)

Build Method	Config Var	Format of Config Var Value
Any	LOOP_APNS_KEY_ID	AAAAAAAAAA
Any	LOOP_APNS_KEY	-----BEGIN PRIVATE KEY----- AA AA AA AAAAAAAA -----END PRIVATE KEY-----
Any	LOOP_DEVELOPER_TEAM_ID	AAAAAAAAAA
Browser ONLY	LOOP_PUSH_SERVER_ENVIRONMENT	production

- If you built the Trio app with Browser Build, you must configure all 4
- If you built the Trio app with Mac-Xcode, you do not include `LOOP_PUSH_SERVER_ENVIRONMENT`

Build *LoopFollow*

This page is under construction. But here's the link:

[Install LoopFollow ↗](#)